

GenAI for Cognitive Deterioration

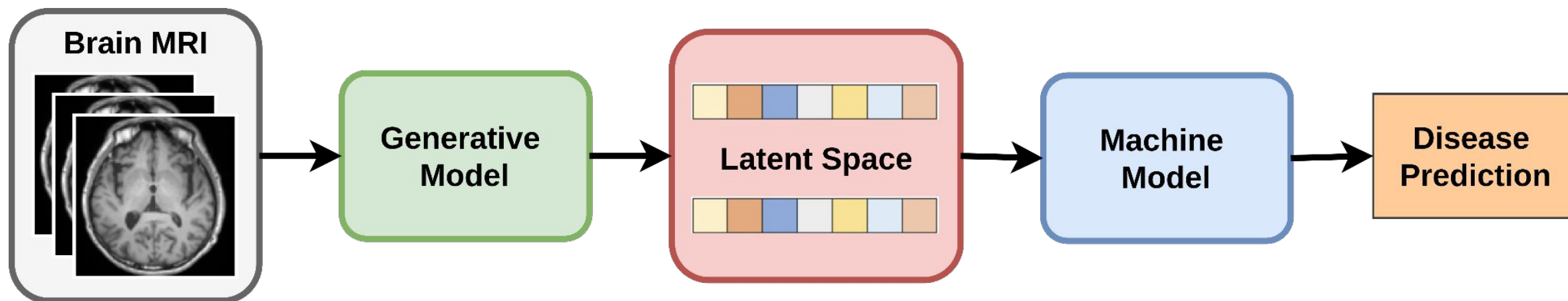
01 | Personal Information



- **Full Name:** Trinh Ngoc Huynh
- **Email:** huynhtn@vnu.edu.vn
- **LinkedIn:** <https://www.linkedin.com/in/huynh-spm/>
- **Github:** <https://github.com/huynhspm>
- **Education:** VNU University of Engineering and Technology (VNU-UET)
 - Bachelor's Degree in Information Technology
 - Second-year Master's Student in Computer Science
- **Experience:**
 - Teacher assistant at the Institute for Artificial Intelligence, VNU-UET
 - Volunteer at AIMS Lab, Vrije Universiteit Brussel (VUB), Brussel, Belgium

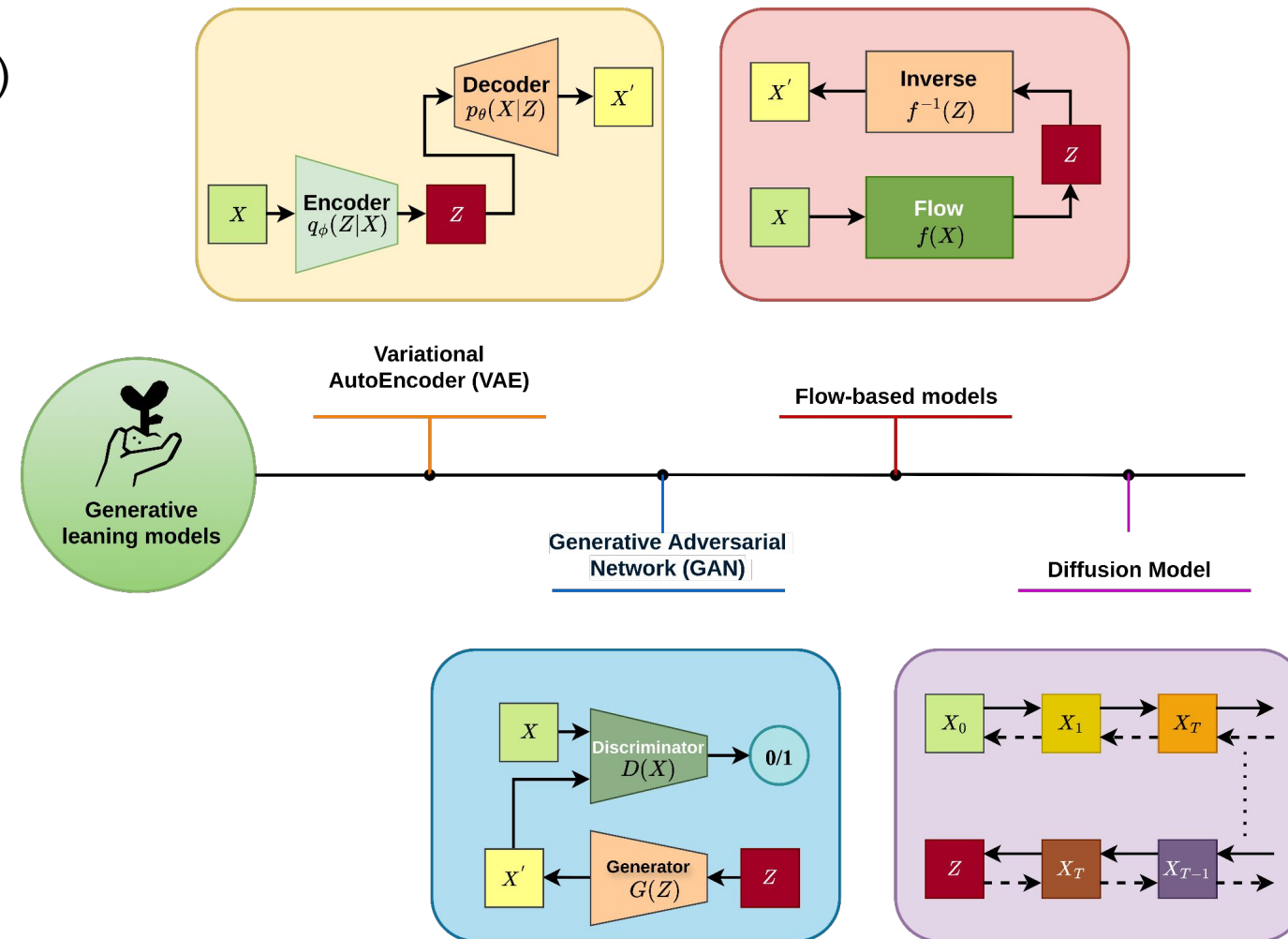
02 | Recent Research Work

- **Research Topic:** Generative AI for Cognitive Impairment Analysis
- Collaboration UET-VUB:
 - Assoc. Prof. Dr. Nguyễn Linh Trung (*UET*)
 - Dr. Trần Quốc Long (*UET*)
 - BSc. Nguyễn Đức Kiên (*UET*)
 - BSc. Trịnh Ngọc Huỳnh (*UET*)
 - Prof. Dr. MD. Guy Nagels (*VUB*)



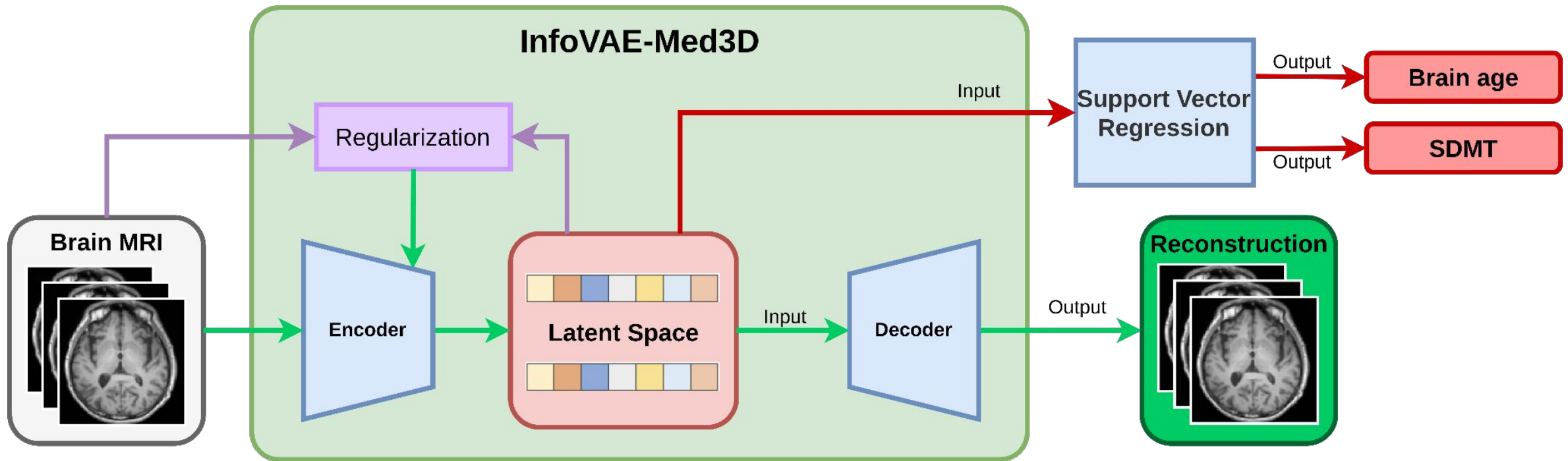
02 | Recent Research Work

- **Research Topic:** Generative AI for Cognitive Impairment Analysis
- Collaboration UET-VUB:
 - Assoc. Prof. Dr. Nguyễn Linh Trung (UET)
 - Dr. Trần Quốc Long (UET)
 - BSc. Nguyễn Đức Kiên (UET)
 - BSc. Trịnh Ngọc Huỳnh (UET)
 - Prof. Dr. MD. Guy Nagels (VUB)



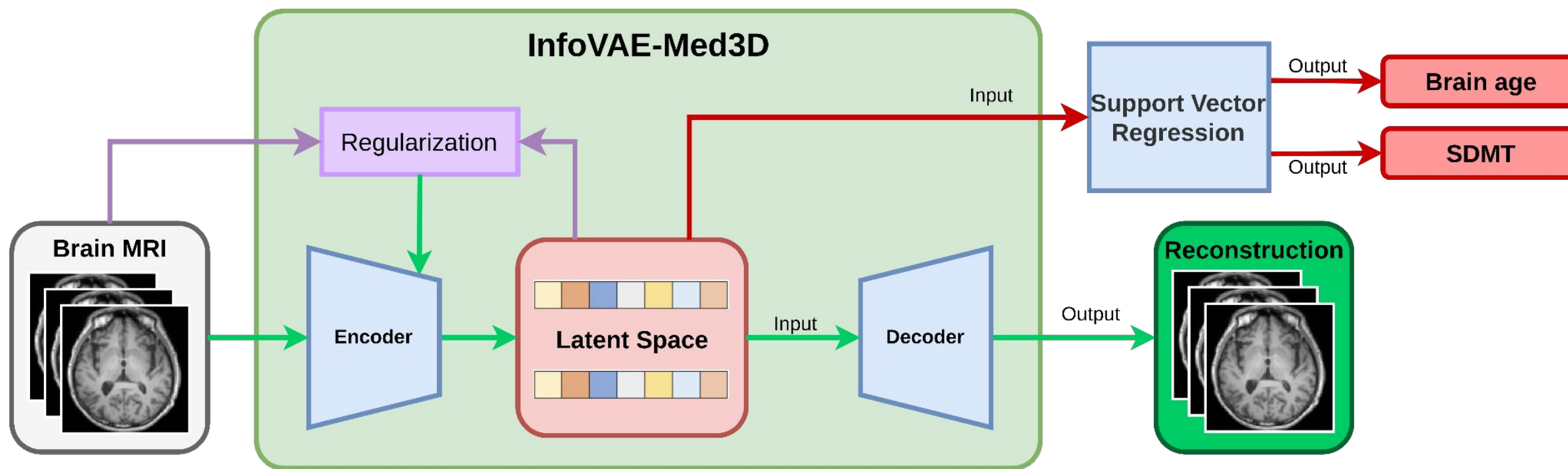
02 | Recent Research Work

- First work with VAE: [Latent Representation for Multiple Sclerosis Prediction](#)
 - **Embedding Model:** InfoVAE-Med3D
 - **Regression Model:** Support Vector Regression (SVR)



02 | Recent Research Work

- First work with VAE: [Latent Representation for Multiple Sclerosis Prediction](#)
 - **Embedding Model:** InfoVAE-Med3D
 - learning structured and lower-dimensional latent representations from 3D brain MRI volumes
 - **Regression Model:** Support Vector Regression (SVR)
 - leveraging these latent vectors for downstream tasks
 - providing both quantitative and qualitative analyses



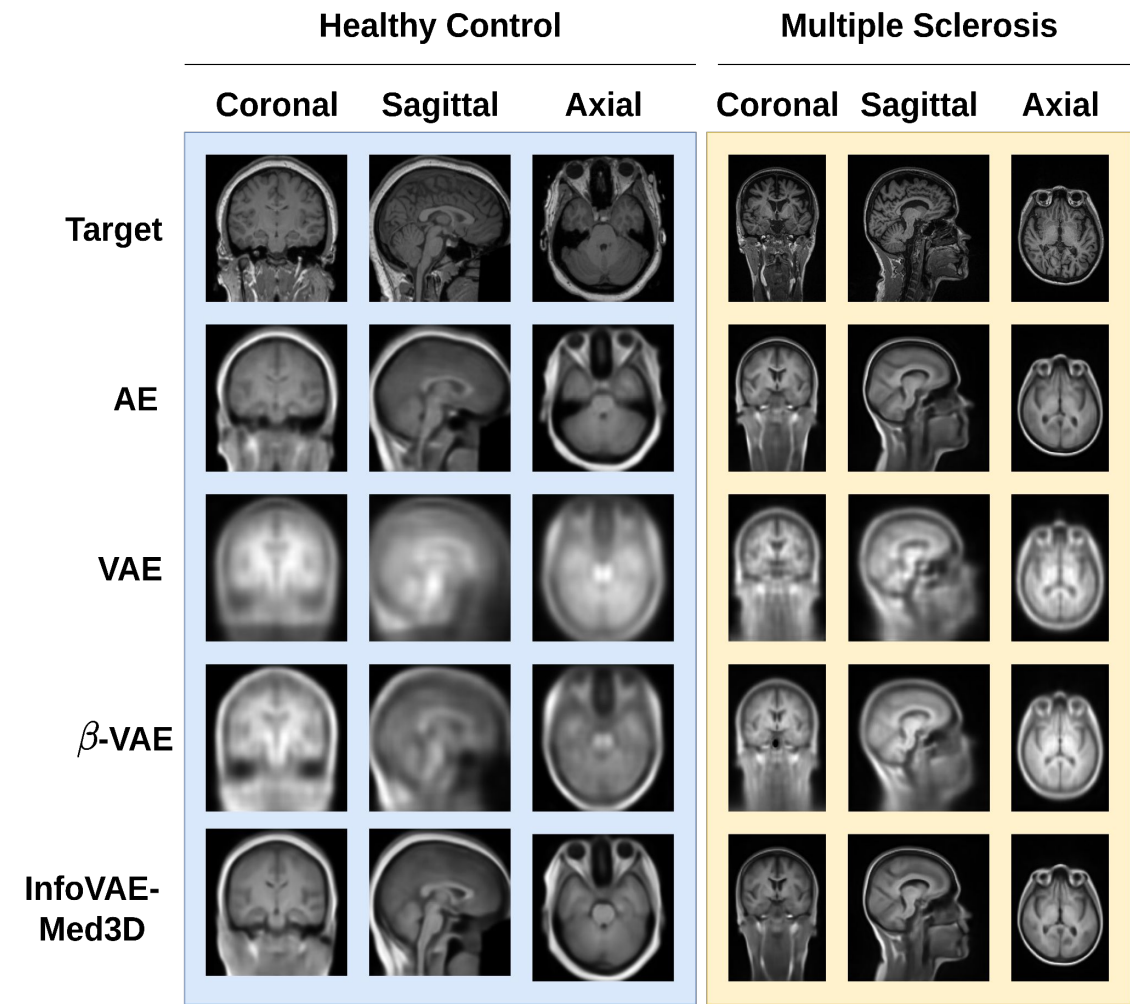
02 | Recent Research Work



- First work with VAE: [Latent Representation for Multiple Sclerosis Prediction](#)
 - Brain-age dataset: public
 - healthy control data
 - chronological age
 - Prague dataset: private
 - MS patient data
 - chronological age + SDMT

Table 1: Reconstruction results on two datasets: our proposed model with multiple configurations compared against three VAE variants. The best results are highlighted in **bold**.

Models	BrainAge		Prague	
	SSIM	PSNR	SSIM	PSNR
AE	0.730	23.93	0.768	24.98
VAE	0.377	19.00	0.616	21.46
β -VAE	0.535	21.17	0.653	23.12
InfoVAE-Med3D ($\alpha = 1, \beta = 1$)	0.519	20.90	0.623	22.97
InfoVAE-Med3D ($\alpha = 0.001, \beta = 1$)	0.554	22.08	0.663	23.65
InfoVAE-Med3D ($\alpha = 0, \beta = 0.1$)	0.741	24.71	0.765	24.97
InfoVAE-Med3D ($\alpha = 0, \beta = 1$)	0.750	24.91	0.789	25.64
InfoVAE-Med3D ($\alpha = 0, \beta = 10$)	0.745	24.76	0.779	25.23



02 | Recent Research Work



- First work with VAE: [Latent Representation for Multiple Sclerosis Prediction](#)
 - Downstream tasks:
 - Brain Age prediction
 - SDMT prediction

Table 2: Quantitative results of downstream tasks on two datasets, comparing the proposed InfoVAE-Med3D with three VAE variants. The best results are highlighted in **bold**.

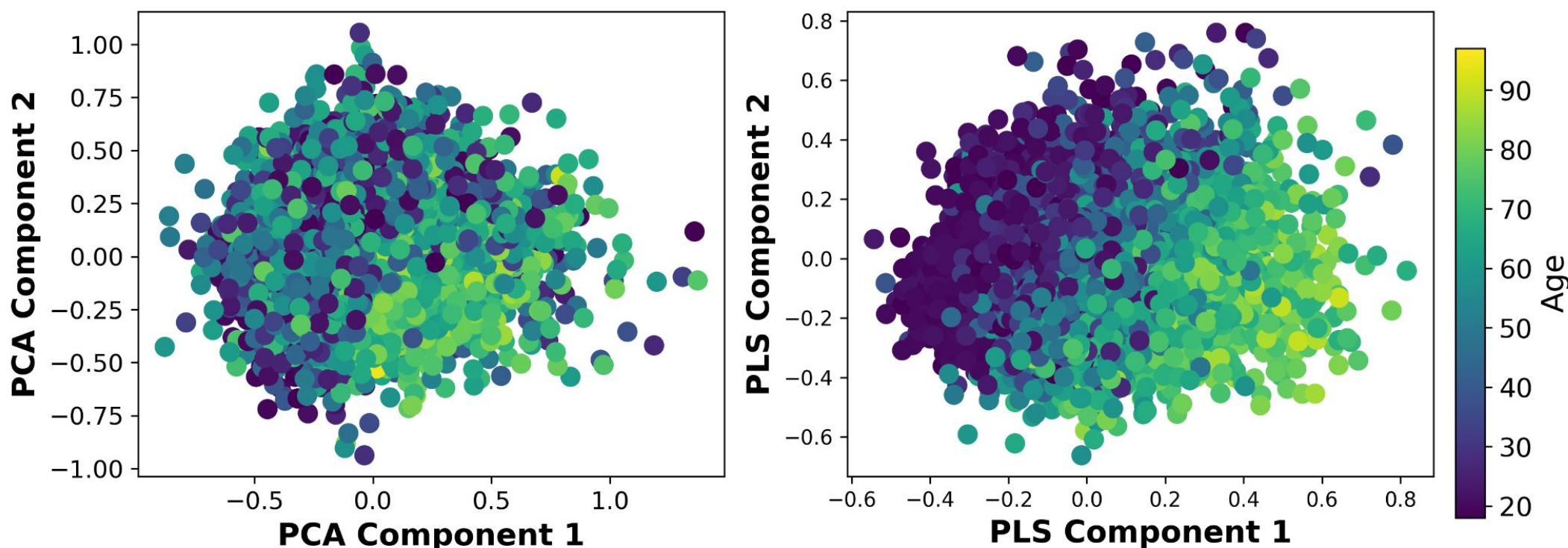
Models	BrainAge			Prague					
	brain age			brain age			SDMT		
	MAE ↓	R^2 ↑	RMSE ↓	MAE ↓	R^2 ↑	RMSE ↓	MAE ↓	R^2 ↑	RMSE ↓
AE	8.348	0.751	10.70	5.233	0.517	6.540	9.005	0.121	11.831
VAE	11.5675	0.425	16.249	7.717	-0.036	9.807	9.711	-0.018	12.35
β -VAE	9.957	0.648	12.820	7.652	-0.016	9.710	9.709	-0.017	12.33
InfoVAE-Med3D	7.664	0.778	10.09	4.736	0.639	5.947	8.531	0.160	10.833

02 | Recent Research Work

- First work with VAE: [Latent Representation for Multiple Sclerosis Prediction](#)

- Downstream tasks: visualization for interpretation
 - Principal Component Analysis (PCA): unsupervised method
 - Partial Least Squares Regression (PLSRegression): supervised method

→ *Latent space shows a smooth transition related to Age from low to high values*

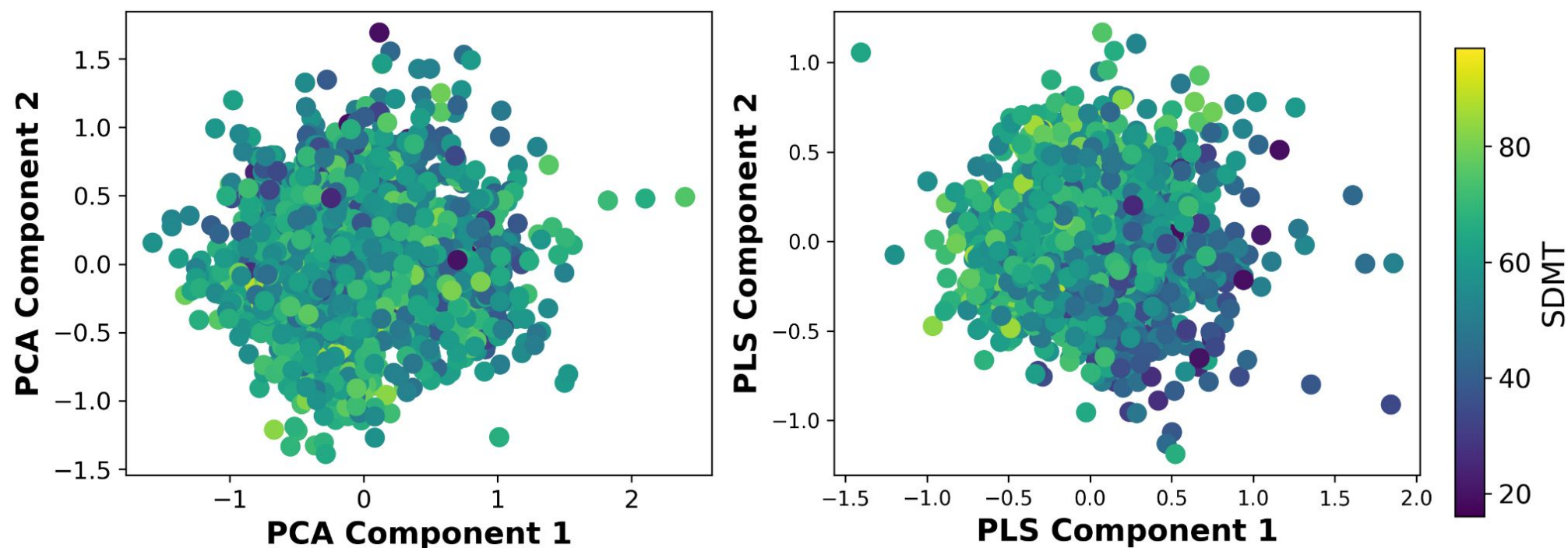


02 | Recent Research Work

- First work with VAE: [Latent Representation for Multiple Sclerosis Prediction](#)

- Downstream tasks: visualization for interpretation
 - Principal Component Analysis (PCA): unsupervised method
 - Partial Least Squares Regression (PLSRegression): supervised method

→ *Latent space shows a smooth transition related to SDMT, but less distinct than age*

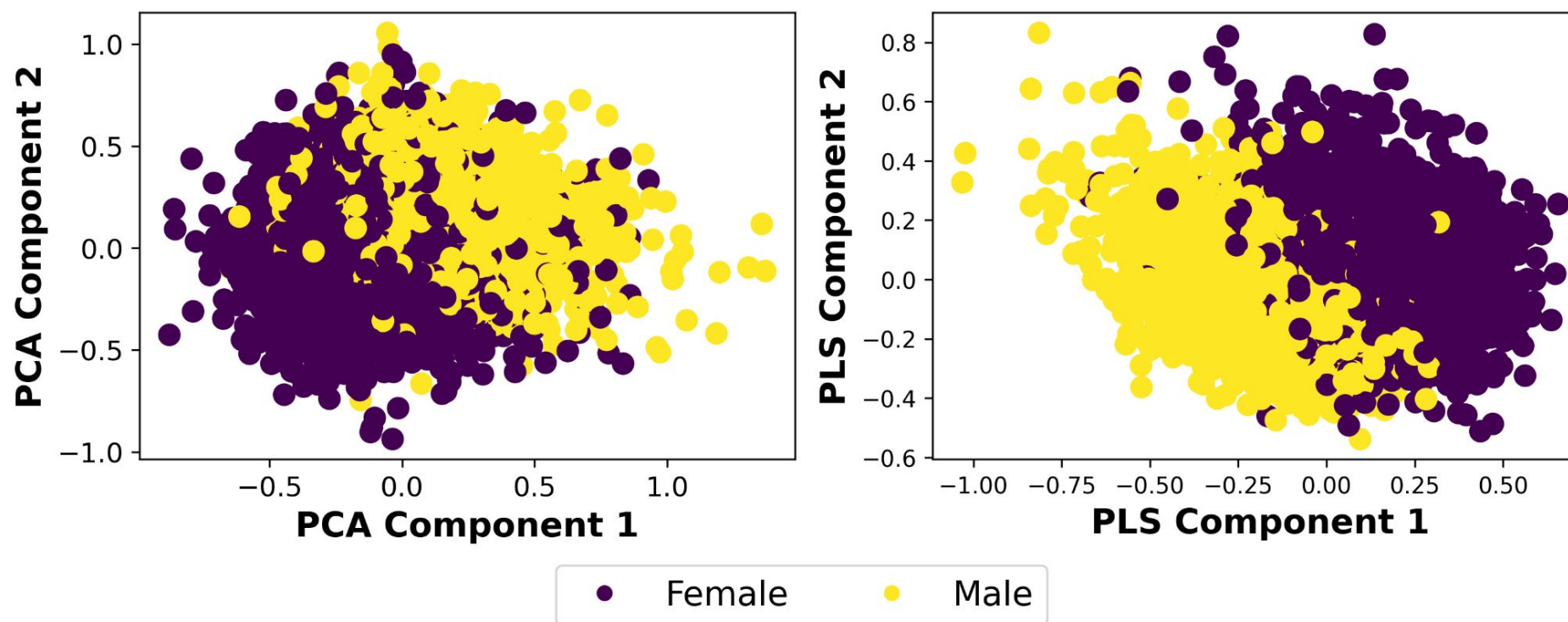


02 | Recent Research Work

- First work with VAE: [Latent Representation for Multiple Sclerosis Prediction](#)

- Downstream tasks: visualization for interpretation
 - Principal Component Analysis (PCA): unsupervised method
 - Partial Least Squares regression (PLSRegression): supervised method

→ *Two most variance dimensions in latent space are strongly correlated with gender information*

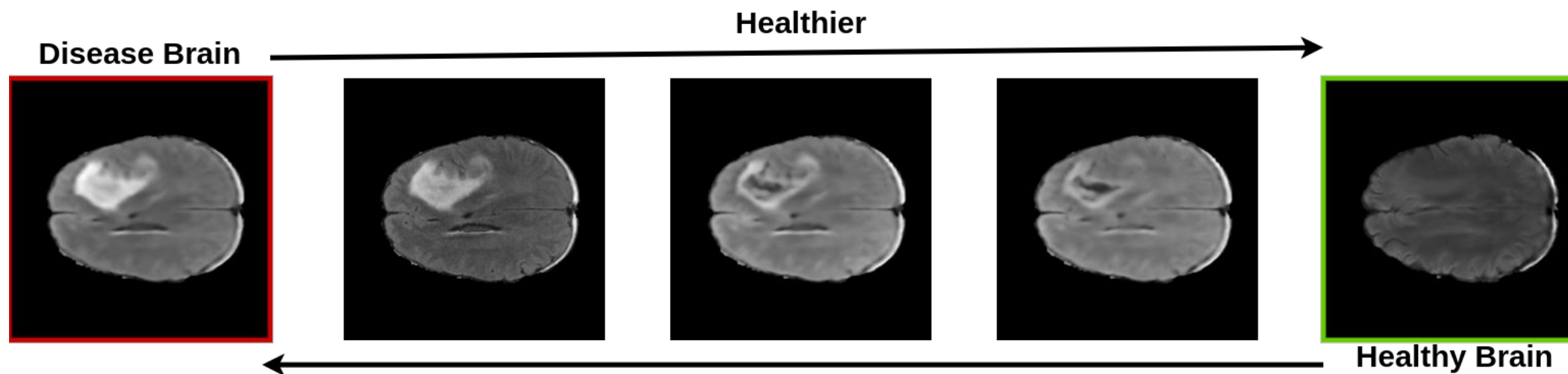


03 | Current Master's Research

- **Research Topic:** Diffusion-based model for Cognitive Impairment Analysis
 - Latent representation learning: learn a ***compact, structured, and informative*** latent vector
 - **Compactness:** high-dimensional MRI volumes are encoded into a low-dimensional latent vector
 - **Structured:** PCA and PLSRegression reveal that the latent space organizes along clinically and biologically meaningful axes.
 - **Informativeness:** the latent retains sufficient information to support
 - Reconstruction + Generation: faithful reconstructions and realistic, diverse synthetic MRI images
 - Regression: accurate prediction of radiological (Brain Age), cognitive (SDMT), and physical disability (EDSS) outcomes using simple machine learning models on the latent vector.

03 | Current Master's Research

- **Research Topic:** Diffusion-based model for Cognitive Impairment Analysis
 - Diseases: Multiple Sclerosis (MS) and NMOSD (Neuromyelitis optica spectrum disorders)
 - Controlling MRI Attributes via Latent Manipulation:
 - Translation from Healthy to Disease Image
 - Translation from MS to NMOSD
- Explore new biomarkers for diseases and difference between MS and NMOSD



Thanks For Listening
